



SRI RAMAKRISHNA
ENGINEERING COLLEGE

[Educational Service : SNR Sons Charitable Trust]
Vattamalaipalayam, N.G.G.O. Colony Post,
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CAPSTONE PROJECT REPORT

STUDENT COURSE REGISTRATION SYSTEM

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THIRD YEAR B.E. CSE – V SEM

Academic Year 2026-2027

20CS279 & WEB TECHNOLOGIES LABORATORY

COURSE COORDINATOR

Mr.N.Manoj,AP/CSE



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STUDENT COURSE REGISTRATION SYSTEM

Assessment

INDEX

S. No.	Name & Roll No.	Introduction (2)	Logic Building (2)	Implementation Steps (4)	Report (2)	Total (10)
1.	AISHWARYA U 71812401006					
2.	AJAY IYANRAJ 71812401007					
3.	AKASHVARMAN R 71812401008					
4.	AKHILESH RAJ P 71812401009					

SIGNATURE OF COURSE COORDINATOR

Mr.N.Manoj,AP/CSE

CONTENTS

S. No.	TITLE	PAGE NUMBER
	STUDENT COURSE REGISTRATION SYSTEM	4
1.	INTRODUCTION	5
2.	LOGIC BUILDING	7
3.	IMPLEMENTATION STEPS	9
4.	CODE AND OUTPUT SCREENSHOTS	11
5.	CONCLUSION	19

STUDENT COURSE REGISTRATION SYSTEM

Abstract

The Student Course Registration System is a web application that allows the students of a college to log in with their roll number, browse the list of courses offered in the semester, register for the courses they want, drop a course, and view the registered courses in a weekly timetable. The system checks every registration against the available seats, the maximum credit limit and the time-slot clash rules, and it shows clear messages when a rule fails. The project is developed with HTML5 for the structure, CSS3 for the design and JavaScript for the logic, and the registrations are stored in the browser localStorage, so the complete application runs on the client side without any server.

Team members

S. No.	Name	Roll Number
1	AISHWARYA U	71812401006
2	AJAY IYANRAJ	71812401007
3	AKASHVARMAN R	71812401008
4	AKHILESH RAJ P	71812401009

Objectives

- To replace the manual, paper based course registration with a simple and fast web application.
- To allow only valid students to log in and to register for courses.
- To prevent over-booking of seats, crossing of the credit limit and clash of class timings.
- To show the registered courses, the total credits and the weekly timetable instantly.
- To keep the design responsive so that it works on desktop and mobile screens.

Technologies used

Technology / Tool	Purpose
HTML5	Page structure: login form, tabs, tables and containers
CSS3	Layout, colours, badges, progress bar, timetable grid and responsive design
JavaScript (ES6)	Login validation, course filtering, registration rules, timetable and localStorage
Browser localStorage	Stores the registrations of all students without a database
Visual Studio Code	Writing and formatting the source code
Google Chrome / Playwright	Running the application and capturing the output screenshots

1. INTRODUCTION

1.1 Background

At the beginning of every semester a student has to choose the courses to study, such as the core subjects, the electives and the laboratory courses. In many colleges this is still done with paper forms or spreadsheets. The staff have to check the seats, the credits and the time-table of every student by hand, which takes a lot of time and often leads to mistakes such as two courses at the same hour or more students than seats.

1.2 Problem Statement

Design and develop a Student Course Registration System in which a student can log in, view the courses with their credits, faculty, time slots and available seats, register or drop courses, and see the total credits and the timetable. The system must automatically reject a registration when the course is full, when the credit limit of 24 is crossed, or when the time slot clashes with an already registered course.

1.3 Scope of the project

- Student side functions: login, search / filter courses, register, drop, view credits and timetable.
- Twelve sample courses of five departments with limited seats and fixed weekly time slots.
- Four sample student accounts (the members of this team) protected by a password.
- A client side implementation only; a real deployment would store the data on a server database.

1.4 Modules of the system

Module	Description
Login	Checks the roll number format, the existence of the student and the password.
Available Courses	Lists all courses with seats left; supports search by code, title, faculty and department filter.
Registration engine	Applies the rules for seats, credits and time clash before saving a registration.
My Registrations	Shows the registered courses, total credits with a progress bar, and a Drop button.
Timetable	Builds a Monday to Friday, five period grid from the slots of the registered courses.
Storage	Saves the registrations of every student in localStorage so they remain after a page refresh.

1.5 Key features

- **Live seat count:** the seats left of every course are updated immediately and shown in green, yellow or red badges.
- **Credit control:** a progress bar shows the credits used out of the maximum of 24.
- **Clash detection:** a course whose slot is already taken is rejected with the name of the clashing course.
- **Responsive design:** the layout adapts to phone screens with a scrollable table.

1.6 Advantages

- Saves the time of both students and the staff.
- Avoids human errors in seat allotment and in time-table clashes.

- Works in any browser without installation, and the data is available offline.

1.7 Existing system versus proposed system

Aspect	Manual / paper based system	Proposed web application
Registration time	Long queues and several days	A few minutes from any device
Seat checking	Counted by the staff by hand	Seats left are calculated automatically
Time-table clash	Found only after registration	Rejected immediately with a message
Credit limit	Verified manually by the advisor	Checked automatically against 24 credits
Errors	Frequent writing and counting mistakes	Rules are enforced by the program
Records	Paper files that can be lost	Saved in the browser and shown instantly

1.8 Users of the system

User	Role	Status in this project
Student	Logs in, searches courses, registers, drops and views the timetable	Implemented
Faculty advisor	Approves the registrations of the students	Future enhancement
Administrator	Creates courses, sets seats and slots	Future enhancement

1.9 Software requirements

Item	Requirement
Operating system	Windows 10 / Linux / macOS
Web browser	Google Chrome, Microsoft Edge, Mozilla Firefox or Safari (latest version)
Editor	Visual Studio Code or any text editor
Internet	Not needed to run the application (static files only)

1.10 Applications

- Semester course registration in engineering and arts colleges.
- Elective selection and add / drop process at the beginning of a semester.
- Online training centres and certificate programmes with limited batch sizes.

1.11 Limitations

- The data is stored in the browser of one computer, so it is not shared between different computers.
- There is no real authentication server; the student list and password are kept in the script for demonstration.
- Only student functions are implemented; faculty and administrator functions are future work.

2. LOGIC BUILDING

2.1 Overall logic

Step 1: Login: The student enters the roll number and the password; the login logic validates both.

Step 2: Show portal: After a successful login the application hides the login view and shows the main view with the student name.

Step 3: List courses: The array of courses is displayed as table rows; the search text and department filter reduce the rows.

Step 4: Validate: When Register is clicked the three rules (seats, credits, clash) are checked one after another.

Step 5: Save: If all rules pass the course code is added to the list of the student and saved in localStorage.

Step 6: Refresh: The seat badges, credit chip, progress bar, My Registrations table and the timetable are redrawn.

2.2 Data used by the program

Each course is stored as a JavaScript object. The registrations of all the students are stored in one object in localStorage, where the key is the roll number and the value is the list of registered course codes.

```
// one course object
{ code: '20CS212', title: 'Web Technologies', dept: 'CSE', credits: 3,
  faculty: 'Mr. N. Manoj', slots: ['Mon-1', 'Wed-2'], seats: 60, taken: 41 }

// registrations saved in localStorage under the key "scrs"
{ '71812401006': ['20CS212', '20CS202'], '71812401007': ['20CS205'] }

// seats left = total seats - seats already taken - registrations of all students
seatsLeft = course.seats - course.taken - countOfStudentsRegistered(course.code)
```

2.3 Business rules

Rule	Condition checked	Message shown
Valid roll number	11 digits and present in the student list	Roll number must contain exactly 11 digits / not found
Valid password	At least 8 characters and equal to the stored password	Incorrect password. Please try again.
Seat availability	seats left > 0	20CS206 is full. No seats left.
Credit limit	registered credits + course credits <= 24	Credit limit of 24 exceeded.
Time clash	no common slot with any registered course	Time clash with 20CS212 at Mon-1.
Duplicate	a registered course shows a disabled button	Registered (button disabled)

2.4 Algorithm for registering a course

```
registerCourse(code):
  course <- find course with this code
  if seatsLeft(course) <= 0                -> show "course is full"; stop
  if totalCredits() + course.credits > 24   -> show "credit limit
exceeded"; stop
  if any registered course shares a slot    -> show "time clash"; stop
  add code to registrations of the current student
```

```

save registrations in localStorage
show "Registered" message
redraw catalogue, my registrations and timetable

```

2.5 Flowchart

Flowchart of login and course registration

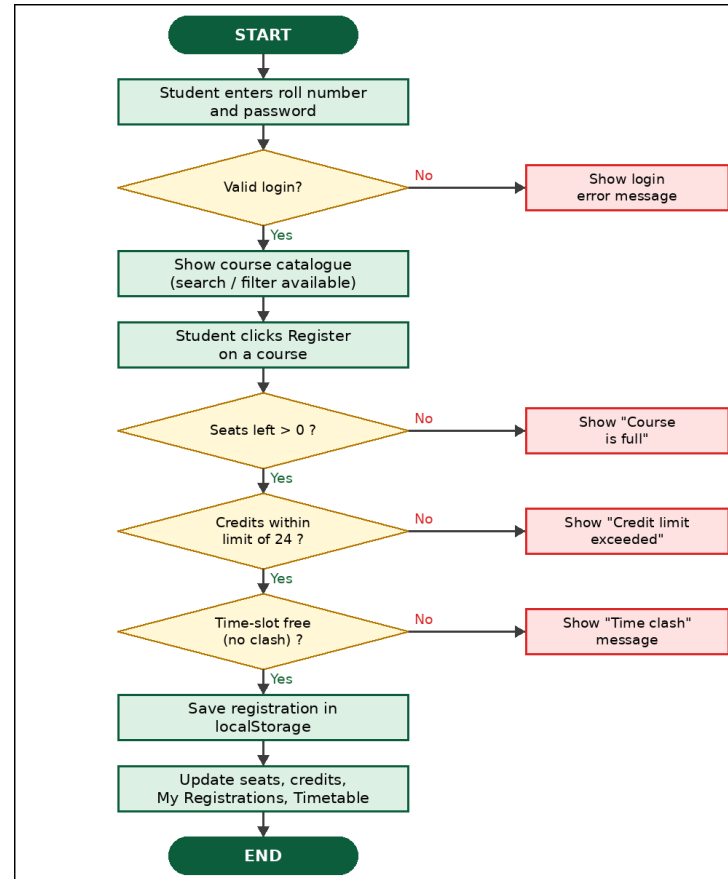


Fig. 1 - Logic of login and course registration.

2.6 Logic of credits and timetable

The total credits are the sum of the credits of the registered courses; the width of the progress bar is $\text{total} / 24 \times 100$ percent. For the timetable every slot is written as Day-Period (for example Wed-2). A map is created from slot to course code and the grid of five days and five periods is drawn by looking up each cell in this map, so a clash is impossible because a slot can hold only one course.

2.7 Example walk-through of the logic

Action of student 71812401006	Rule checked	Result
Register 20CS212 (3 credits, Mon-1, Wed-2)	Seats, credits $3 \leq 24$, no clash	Registered; credits 3 / 24
Register 20CS202 (3 credits, Mon-3, Fri-2)	Seats, credits $6 \leq 24$, no clash	Registered; credits 6 / 24
Register 20CS204 (Mon-1, Thu-3)	Slot Mon-1 already used by 20CS212	Rejected: time clash
Register 20CS206 (4 credits)	Seats left = 0	Rejected: course is full
Drop 20CS202	Course removed from the list	Credits return to 3 / 24
Open Timetable	Slots mapped to periods	Mon-1 and Wed-2 highlighted

3. IMPLEMENTATION STEPS

3.1 Steps followed

Step 1: Create the project folder with three files: index.html, style.css and script.js.

Step 2: Design the HTML: a login section, and a main section with a top bar, three tabs and three panels.

Step 3: Style with CSS: green theme, cards, badges, progress bar, timetable grid and a media query for phones.

Step 4: Store the data: define the student list, the constants and the array of twelve courses in script.js.

Step 5: Write the login logic: validate the roll number and the password and switch the views.

Step 6: Display the catalogue: build the table rows with template strings; add the search box and the department filter.

Step 7: Write the registration engine: functions seatsLeft, totalCredits, findClash, registerCourse and dropCourse.

Step 8: Build My Registrations and Timetable: render the list, the credit bar and the weekly grid.

Step 9: Save to localStorage after every register or drop so the data survives a refresh.

Step 10: Test the application with wrong login, search, filter, registration, clash, full course and drop cases and capture the screenshots.

3.2 Project files

File	Lines	Purpose
index.html	84	Structure of the login page, the tabs, the tables and the toast message box
style.css	66	Appearance of every component and the responsive rules
script.js	171	Data, login, filtering, registration rules, timetable and storage

3.3 Main functions of script.js

Function	Purpose
seatsLeft(course)	Returns the seats still free after counting the registrations of all students
totalCredits()	Adds the credits of the courses registered by the current student
findClash(course)	Returns the registered course that shares a time slot with the given course
registerCourse(code)	Applies seat, credit and clash rules and saves the registration
dropCourse(code)	Removes a registered course and refreshes the screen
renderCatalogue()	Draws the course table after applying the search text and department filter
renderMine()	Draws the registered courses and the credit progress bar
renderTimetable()	Draws the Monday to Friday timetable grid
toast(msg, type)	Shows a green success or red error message for a short time

3.4 Test cases

S. No.	Test case	Expected result	Result
1	Login with a wrong password	Error message "Incorrect password"	Pass
2	Login with a valid roll number and password	Portal opens with the student name	Pass
3	Filter by department AI&DS	Only the two AI&DS courses are listed	Pass
4	Register 20CS212	Success message and credits become 3 / 24	Pass
5	Register 20CS204 after 20CS212	Error: time clash at Mon-1	Pass
6	Open a full course (20CS206)	Button shows Full and is disabled	Pass
7	Open the Timetable tab	Registered slots are highlighted	Pass

3.5 Folder structure

```
student-course-registration/  
|-- index.html      (structure of the pages)  
|-- style.css       (design and responsive rules)  
|-- script.js       (data, logic and localStorage)  
|-- shots/          (output screenshots used in this report)
```

3.6 How to run the application

Step 1: Copy the three files index.html, style.css and script.js into one folder.

Step 2: Double-click index.html; it opens in the default browser and shows the login page.

Step 3: Log in with one of the sample accounts given below.

Step 4: Register courses, open the other tabs and use Logout when finished.

3.7 Sample student accounts

S. No.	Name	Roll number (user name)	Password
1	AISHWARYA U	71812401006	Srec@2026
2	AJAY IYANRAJ	71812401007	Srec@2026
3	AKASHVARMAN R	71812401008	Srec@2026
4	AKHILESH RAJ P	71812401009	Srec@2026

3.8 Coding conventions followed

- Meaningful names for functions and variables (registerCourse, seatsLeft, totalCredits).
- Data (courses, students) is kept separate from the display and rule functions.
- Comments mark every section of the script: data, storage, login, tabs, catalogue, register / drop, timetable.
- One function performs one job, so each rule can be tested separately.
- Constants such as MAX_CREDITS are declared once and reused.

4. CODE AND OUTPUT SCREENSHOTS

4.1 index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Student Course Registration System</title>
<link rel="stylesheet" href="style.css">
</head>
<body>

<!-- ===== LOGIN VIEW ===== -->
<section id="loginView" class="login-wrap">
  <form id="loginForm" class="card login-card" novalidate>
    <h1>Student Course Registration</h1>
    <p class="sub">Sri Ramakrishna Engineering College</p>
    <label for="roll">Roll Number</label>
    <input type="text" id="roll" placeholder="e.g. 71812401006" maxlength="11"
inputmode="numeric" required>
    <label for="pwd">Password</label>
    <input type="password" id="pwd" placeholder="Enter password" required>
    <div id="loginError" class="error"></div>
    <button type="submit" class="btn primary block">Login</button>
    <p class="hint">Demo password for all students: <b>Srec@2026</b></p>
  </form>
</section>

<!-- ===== MAIN APPLICATION ===== -->
<section id="appView" class="hidden">
  <header class="topbar">
    <div>
      <h2>Course Registration Portal</h2>
      <span id="welcome"></span>
    </div>
    <div class="right">
      <span class="chip" id="creditChip"></span>
      <button id="logoutBtn" class="btn light">Logout</button>
    </div>
  </header>

  <nav class="tabs">
    <button class="tab active" data-tab="catalogue">Available Courses</button>
    <button class="tab" data-tab="mine">My Registrations</button>
    <button class="tab" data-tab="timetable">Timetable</button>
  </nav>

  <main>
    <div id="catalogue" class="panel">
      <div class="filters">
        <input type="search" id="search" placeholder="Search by code, title or faculty">
        <select id="deptFilter">
          <option value="All">All Departments</option>
        </select>
      </div>
      <table>
        <thead>
          <tr><th>Code</th><th>Course Title</th><th>Dept</th><th>Credits</th>
            <th>Faculty</th><th>Slots</th><th>Seats Left</th><th>Action</th></tr>
        </thead>
        <tbody id="courseBody"></tbody>
      </table>
    </div>

    <div id="mine" class="panel hidden">
      <div class="progress-box">
        <span id="creditText"></span>
        <div class="bar"><div id="creditBar"></div></div>
      </div>
      <table>
        <thead>
          <tr><th>Code</th><th>Course
Title</th><th>Credits</th><th>Faculty</th><th>Slots</th><th>Action</th></tr>
        </thead>
        <tbody id="myBody"></tbody>
      </table>
    </div>
  </main>
</section>
```

```

        <div id="timetable" class="panel hidden">
            <table class="grid" id="ttTable"></table>
        </div>
    </main>
</section>

<div id="toast" class="toast hidden"></div>
<script src="script.js"></script>
</body>
</html>

```

4.2 style.css

```

* { box-sizing: border-box; }
body { margin: 0; font-family: "Segoe UI", Arial, sans-serif; background: #eef2f7; color: #1f2937; }
.hidden { display: none !important; }

/* login */
.login-wrap { min-height: 100vh; display: flex; align-items: center; justify-content: center; background: linear-gradient(135deg, #0b5d3b, #1f8a5b); padding: 16px; }
.card { background: #fff; border-radius: 12px; box-shadow: 0 6px 24px rgba(0,0,0,.25); }
.login-card { width: 100%; max-width: 380px; padding: 28px; }
.login-card h1 { margin: 0; font-size: 22px; color: #0b5d3b; text-align: center; }
.sub { text-align: center; color: #64748b; margin: 4px 0 20px; font-size: 14px; }
label { display: block; font-weight: 600; font-size: 14px; margin: 12px 0 5px; }
input, select { width: 100%; padding: 10px; font-size: 14px; border: 1px solid #94a3b8; border-radius: 6px; }
input:focus, select:focus { outline: 2px solid #34a56f; }
.error { color: #dc2626; font-size: 13px; min-height: 18px; margin-top: 6px; }
.hint { font-size: 12px; color: #64748b; text-align: center; margin-bottom: 0; }

/* buttons */
.btn { padding: 8px 16px; border: 0; border-radius: 6px; cursor: pointer; font-size: 14px; }
.btn.primary { background: #0b5d3b; color: #fff; }
.btn.light { background: #fff; color: #0b5d3b; }
.btn.danger { background: #dc2626; color: #fff; }
.btn.block { width: 100%; margin-top: 8px; padding: 11px; }
.btn.disabled { background: #cbd5e1; color: #64748b; cursor: not-allowed; }

/* top bar and tabs */
.topbar { display: flex; justify-content: space-between; align-items: center; flex-wrap: wrap; gap: 8px; background: #0b5d3b; color: #fff; padding: 14px 24px; }
.topbar h2 { margin: 0; font-size: 20px; }
.topbar span#welcome { font-size: 13px; opacity: .9; }
.right { display: flex; align-items: center; gap: 12px; }
.chip { background: rgba(255,255,255,.2); padding: 6px 12px; border-radius: 20px; font-size: 13px; }
.tabs { display: flex; gap: 4px; padding: 12px 24px 0; background: #fff; border-bottom: 2px solid #0b5d3b; }
.tab { border: 0; background: #e2e8f0; padding: 10px 18px; border-radius: 8px 8px 0 0; cursor: pointer; font-size: 14px; }
.tab.active { background: #0b5d3b; color: #fff; }
main { max-width: 1100px; margin: 20px auto; padding: 0 16px; }

/* tables */
.filters { display: flex; gap: 12px; margin-bottom: 14px; }
.filters input { flex: 2; }
.filters select { flex: 1; }
table { width: 100%; border-collapse: collapse; background: #fff; border-radius: 8px; overflow: hidden; box-shadow: 0 1px 6px rgba(0,0,0,.12); font-size: 14px; }
th { background: #0b5d3b; color: #fff; text-align: left; padding: 10px; }
td { padding: 10px; border-bottom: 1px solid #e2e8f0; }
tr:hover td { background: #f0fdf4; }
.badge { padding: 3px 9px; border-radius: 12px; font-size: 12px; font-weight: 600; }
.badge.ok { background: #dcfce7; color: #166534; }
.badge.low { background: #fef3c7; color: #92400e; }
.badge.full { background: #fee2e2; color: #991b1b; }

/* credit progress and timetable */
.progress-box { background: #fff; padding: 14px; border-radius: 8px; margin-bottom: 14px; box-shadow: 0 1px 6px rgba(0,0,0,.12); }
.bar { height: 12px; background: #e2e8f0; border-radius: 6px; margin-top: 8px; overflow: hidden; }
.bar div { height: 100%; width: 0; background: #16a34a; transition: width .4s; }
.grid th, .grid td { text-align: center; border: 1px solid #cbd5e1; height: 56px; }
.grid td.filled { background: #dcfce7; font-weight: 600; font-size: 13px; }

```

```

/* toast */
.toast { position: fixed; bottom: 24px; right: 24px; padding: 12px 20px; border-radius: 8px;
color: #fff;
    box-shadow: 0 4px 14px rgba(0,0,0,.3); font-size: 14px; max-width: 360px; }
.toast.success { background: #16a34a; } .toast.error { background: #dc2626; }

@media (max-width: 700px) {
    .filters { flex-direction: column; }
    table { display: block; overflow-x: auto; white-space: nowrap; }
    .topbar { padding: 12px 14px; }
}

```

4.3 script.js

```

// ----- DATA -----
const PASSWORD = 'Srec@2026';
const STUDENTS = {
    '71812401006': 'Aishwarya U',
    '71812401007': 'Ajay Iyanraj',
    '71812401008': 'Akashvarman R',
    '71812401009': 'Akhilesh Raj P'
};
const MAX_CREDITS = 24;
const DAYS = ['Mon', 'Tue', 'Wed', 'Thu', 'Fri'];
const PERIODS = ['9:00 - 10:00', '10:00 - 11:00', '11:15 - 12:15', '1:15 - 2:15', '2:15 - 3:15'];

// seats = total seats, taken = seats already occupied by other students
const COURSES = [
    { code: '20CS212', title: 'Web Technologies', dept: 'CSE', credits: 3, faculty: 'Mr. N. Manoj', slots: ['Mon-1', 'Wed-2'], seats: 60, taken: 41 },
    { code: '20CS201', title: 'Data Structures', dept: 'CSE', credits: 4, faculty: 'Dr. S. Priya', slots: ['Tue-1', 'Thu-1'], seats: 60, taken: 52 },
    { code: '20CS202', title: 'Database Management', dept: 'CSE', credits: 3, faculty: 'Ms. R. Kavitha', slots: ['Mon-3', 'Fri-2'], seats: 60, taken: 38 },
    { code: '20CS203', title: 'Operating Systems', dept: 'CSE', credits: 3, faculty: 'Dr. M. Suresh', slots: ['Tue-2', 'Fri-1'], seats: 60, taken: 59 },
    { code: '20CS204', title: 'Computer Networks', dept: 'CSE', credits: 3, faculty: 'Mr. K. Arun', slots: ['Mon-1', 'Thu-3'], seats: 60, taken: 30 },
    { code: '20CS205', title: 'Software Engineering', dept: 'CSE', credits: 3, faculty: 'Ms. L. Divya', slots: ['Wed-3', 'Fri-4'], seats: 60, taken: 45 },
    { code: '20CS206', title: 'Artificial Intelligence', dept: 'AI&DS', credits: 4, faculty: 'Dr. V. Rajesh', slots: ['Tue-4', 'Thu-2'], seats: 40, taken: 40 },
    { code: '20CS207', title: 'Cyber Security', dept: 'AI&DS', credits: 3, faculty: 'Mr. T. Harish', slots: ['Wed-4', 'Fri-3'], seats: 40, taken: 22 },
    { code: '20CS208', title: 'Cloud Computing', dept: 'IT', credits: 3, faculty: 'Ms. P. Anitha', slots: ['Thu-4', 'Tue-3'], seats: 50, taken: 35 },
    { code: '20MA201', title: 'Discrete Mathematics', dept: 'MATHS', credits: 4, faculty: 'Dr. G. Latha', slots: ['Mon-2', 'Wed-1'], seats: 70, taken: 48 },
    { code: '20HS201', title: 'Professional Ethics', dept: 'HSS', credits: 2, faculty: 'Mr. B. Karthik', slots: ['Fri-5'], seats: 80, taken: 50 },
    { code: '20CS279', title: 'Web Technologies Lab', dept: 'CSE', credits: 2, faculty: 'Mr. N. Manoj', slots: ['Thu-5'], seats: 60, taken: 44 }
];

// ----- STORAGE (registrations of all students) -----
let currentUser = null;
const store = {
    load() { try { return JSON.parse(localStorage.getItem('scrs')) || {}; } catch (e) { return {}; } },
    save(data) { localStorage.setItem('scrs', JSON.stringify(data)); }
};
const $ = id => document.getElementById(id);

function myCodes() { return store.load()[currentUser] || []; }
function courseByCode(code) { return COURSES.find(c => c.code === code); }
function seatsLeft(course) {
    const all = store.load();
    const mine = Object.values(all).filter(list => list.includes(course.code)).length;
    return course.seats - course.taken - mine;
}
function totalCredits() { return myCodes().reduce((sum, c) => sum + courseByCode(c).credits, 0); }

// ----- TOAST -----
let toastTimer;
function toast(msg, type) {
    const t = $('toast');

```

```

t.textContent = msg; t.className = 'toast ' + type;
clearTimeout(toastTimer);
toastTimer = setTimeout(() => t.classList.add('hidden'), 3500);
}

// ----- LOGIN / LOGOUT -----
$('loginForm').addEventListener('submit', e => {
  e.preventDefault();
  const roll = $('roll').value.trim(), pwd = $('pwd').value;
  let msg = '';
  if (!/^\d{11}$/.test(roll)) msg = 'Roll number must contain exactly 11 digits.';
  else if (!STUDENTS[roll]) msg = 'Roll number not found.';
  else if (pwd.length < 8) msg = 'Password must be at least 8 characters.';
  else if (pwd !== PASSWORD) msg = 'Incorrect password. Please try again.';
  $('loginError').textContent = msg;
  if (msg) return;
  currentUser = roll;
  $('loginView').classList.add('hidden');
  $('appView').classList.remove('hidden');
  $('welcome').textContent = 'Welcome, ' + STUDENTS[roll] + ' (' + roll + ')';
  refreshAll();
});

$('logoutBtn').addEventListener('click', () => {
  currentUser = null;
  $('appView').classList.add('hidden');
  $('loginView').classList.remove('hidden');
  $('loginForm').reset();
});

// ----- TABS -----
document.querySelectorAll('.tab').forEach(btn => btn.addEventListener('click', () => {
  document.querySelectorAll('.tab').forEach(b => b.classList.toggle('active', b === btn));
  document.querySelectorAll('.panel').forEach(p => p.classList.toggle('hidden', p.id !==
  btn.dataset.tab));
}));

// ----- CATALOGUE -----
[...new Set(COURSES.map(c => c.dept))].forEach(d => {
  const o = document.createElement('option'); o.value = o.textContent = d; $
  ('deptFilter').appendChild(o);
});

function renderCatalogue() {
  const q = $('search').value.toLowerCase(), dept = $('deptFilter').value;
  const rows = COURSES.filter(c =>
    (dept === 'All' || c.dept === dept) &&
    (c.code + c.title + c.faculty).toLowerCase().includes(q));
  $('courseBody').innerHTML = rows.map(c => {
    const left = seatsLeft(c), done = myCodes().includes(c.code);
    const cls = left <= 0 ? 'full' : left <= 10 ? 'low' : 'ok';
    const btn = done ? '<button class="btn" disabled>Registered</button>'
      : left <= 0 ? '<button class="btn" disabled>Full</button>'
      : '<button class="btn primary" onclick="registerCourse(\'' + c.code +
  '\')">Register</button>';
    return '<tr><td>' + c.code + '</td><td>' + c.title + '</td><td>' + c.dept + '</td><td>' +
  c.credits +
    '</td><td>' + c.faculty + '</td><td>' + c.slots.join(', ') + '</td><td><span
  class="badge ' + cls + '">' +
    Math.max(left, 0) + '</span></td><td>' + btn + '</td></tr>';
  }).join('') || '<tr><td colspan="8">No courses match your search.</td></tr>';
}
$('search').addEventListener('input', renderCatalogue);
$('deptFilter').addEventListener('change', renderCatalogue);

// ----- REGISTER / DROP -----
function findClash(course) {
  for (const code of myCodes()) {
    const other = courseByCode(code);
    const shared = other.slots.find(s => course.slots.includes(s));
    if (shared) return { other, slot: shared };
  }
  return null;
}

function registerCourse(code) {
  const course = courseByCode(code), all = store.load();
  const clash = findClash(course);
  if (seatsLeft(course) <= 0) return toast(course.code + ' is full. No seats left.', 'error');
  if (totalCredits() + course.credits > MAX_CREDITS)
    return toast('Credit limit of ' + MAX_CREDITS + ' exceeded.', 'error');
}

```

```

    if (clash) return toast('Time clash with ' + clash.other.code + ' at ' + clash.slot + '.',
    'error');
    all[currentUser] = [...myCodes(), code];
    store.save(all);
    toast('Registered for ' + course.code + ' - ' + course.title, 'success');
    refreshAll();
  }

function dropCourse(code) {
  const all = store.load();
  all[currentUser] = myCodes().filter(c => c !== code);
  store.save(all);
  toast('Dropped ' + code, 'success');
  refreshAll();
}

// ----- MY REGISTRATIONS -----
function renderMine() {
  const list = myCodes().map(courseByCode), total = totalCredits();
  $('creditText').textContent = 'Credits registered: ' + total + ' / ' + MAX_CREDITS +
  ' | Courses: ' + list.length;
  $('creditBar').style.width = (total / MAX_CREDITS * 100) + '%';
  $('myBody').innerHTML = list.map(c =>
    '<tr><td>' + c.code + '</td><td>' + c.title + '</td><td>' + c.credits + '</td><td>' +
    c.faculty +
    '</td><td>' + c.slots.join(', ') + '</td><td><button class="btn danger"
    onclick="dropCourse(\'' +
    c.code + '\')">Drop</button></td></tr>').join('') ||
    '<tr><td colspan="6">You have not registered for any course yet.</td></tr>';
  }

// ----- TIMETABLE -----
function renderTimetable() {
  const slotMap = {};
  myCodes().map(courseByCode).forEach(c => c.slots.forEach(s => slotMap[s] = c.code));
  let html = '<tr><th>Period</th>' + DAYS.map(d => '<th>' + d + '</th>').join('') + '</tr>';
  PERIODS.forEach((time, p) => {
    html += '<tr><th>' + time + '</th>' + DAYS.map(d => {
      const code = slotMap[d + '-' + (p + 1)];
      return code ? '<td class="filled">' + code + '</td>' : '<td></td>';
    }).join('') + '</tr>';
  });
  $('ttTable').innerHTML = html;
}

function refreshAll() {
  $('creditChip').textContent = 'Credits: ' + totalCredits() + ' / ' + MAX_CREDITS;
  renderCatalogue(); renderMine(); renderTimetable();
}

```

4.4 Output screenshots

The screenshots below were captured from the running application in a Chromium browser.

Fig. 2 - Login page asking for the roll number and password.

Student Course Registration

Sri Ramakrishna Engineering College

Roll Number

Password

Incorrect password. Please try again.

Login

Demo password for all students: Srec@2026

Fig. 3 - Error message shown for an incorrect password.

Course Registration Portal							
Welcome, Aishwarya U (71812401006)						Credits: 0 / 24	Logout
Available Courses My Registrations Timetable							
Search by code, title or faculty					All Departments		
Code	Course Title	Dept	Credits	Faculty	Slots	Seats Left	Action
20CS212	Web Technologies	CSE	3	Mr. N. Manoj	Mon-1, Wed-2	19	Register
20CS201	Data Structures	CSE	4	Dr. S. Priya	Tue-1, Thu-1	8	Register
20CS202	Database Management	CSE	3	Ms. R. Kavitha	Mon-3, Fri-2	22	Register
20CS203	Operating Systems	CSE	3	Dr. M. Suresh	Tue-2, Fri-1	1	Register
20CS204	Computer Networks	CSE	3	Mr. K. Arun	Mon-1, Thu-3	30	Register
20CS205	Software Engineering	CSE	3	Ms. L. Divya	Wed-3, Fri-4	15	Register
20CS206	Artificial Intelligence	AI&DS	4	Dr. V. Rajesh	Tue-4, Thu-2	0	Full
20CS207	Cyber Security	AI&DS	3	Mr. T. Harish	Wed-4, Fri-3	18	Register
20CS208	Cloud Computing	IT	3	Ms. P. Anitha	Thu-4, Tue-3	15	Register
20MA201	Discrete Mathematics	MATHS	4	Dr. G. Latha	Mon-2, Wed-1	22	Register
20HS201	Professional Ethics	HSS	2	Mr. B. Karthik	Fri-5	30	Register
20CS279	Web Technologies Lab	CSE	2	Mr. N. Manoj	Thu-5	16	Register

Fig. 4 - Course catalogue with credits, faculty, slots and seat badges (20CS206 is full).

Course Registration Portal							
Welcome, Aishwarya U (71812401006)						Credits: 0 / 24	Logout
Available Courses My Registrations Timetable							
Search by code, title or faculty					AI&DS		
Code	Course Title	Dept	Credits	Faculty	Slots	Seats Left	Action
20CS206	Artificial Intelligence	AI&DS	4	Dr. V. Rajesh	Tue-4, Thu-2	0	Full
20CS207	Cyber Security	AI&DS	3	Mr. T. Harish	Wed-4, Fri-3	18	Register

Fig. 5 - Filtering by the AI&DS department shows only two courses.

Course Registration Portal

Welcome, Aishwarya U (71812401006)

Credits: 0 / 24

Logout

Available Courses

My Registrations

Timetable

web

All Departments

Code	Course Title	Dept	Credits	Faculty	Slots	Seats Left	Action
20CS212	Web Technologies	CSE	3	Mr. N. Manoj	Mon-1, Wed-2	19	Register
20CS279	Web Technologies Lab	CSE	2	Mr. N. Manoj	Thu-5	16	Register

Fig. 6 - Searching "web" lists Web Technologies and Web Technologies Lab.

Course Registration Portal

Welcome, Aishwarya U (71812401006)

Credits: 3 / 24

Logout

Available Courses

My Registrations

Timetable

Search by code, title or faculty

All Departments

Code	Course Title	Dept	Credits	Faculty	Slots	Seats Left	Action
20CS212	Web Technologies	CSE	3	Mr. N. Manoj	Mon-1, Wed-2	18	Registered
20CS201	Data Structures	CSE	4	Dr. S. Priya	Tue-1, Thu-1	8	Register
20CS202	Database Management	CSE	3	Ms. R. Kavitha	Mon-3, Fri-2	22	Register
20CS203	Operating Systems	CSE	3	Dr. M. Suresh	Tue-2, Fri-1	1	Deny
20CS204	Computer Networks	CSE	3	Mr. K. Arun	Mon-1, Thu-3	30	Register
20CS205	Software Engineering	CSE	3	Ms. L. Divya	Wed-3, Fri-4	15	Register
20CS206	Artificial Intelligence	AI&DS	4	Dr. V. Rajesh	Tue-4, Thu-2	0	Full
20CS207	Cyber Security	AI&DS	3	Mr. T. Harish	Wed-4, Fri-3	18	Register
20CS208	Cloud Computing	IT	3	Ms. P. Anitha	Thu-4, Tue-3	15	Register
20MA201	Discrete Mathematics	MATHS	4	Dr. G. Latha	Mon-2, Wed-1	22	Register
20HS201	Professional Ethics	HSS	2	Mr. B. Karthik	Fri-5	30	Register
20CS279	Web Technologies Lab	CSE	2	Mr. N. Manoj	Thu-5	16	Register

Fig. 7 - Registering 20CS212 shows a green message and the button changes to Registered.

Course Registration Portal

Welcome, Aishwarya U (71812401006)

Credits: 12 / 24

Logout

Available Courses

My Registrations

Timetable

Search by code, title or faculty

All Departments

Code	Course Title	Dept	Credits	Faculty	Slots	Seats Left	Action
20CS212	Web Technologies	CSE	3	Mr. N. Manoj	Mon-1, Wed-2	18	Registered
20CS201	Data Structures	CSE	4	Dr. S. Priya	Tue-1, Thu-1	8	Register
20CS202	Database Management	CSE	3	Ms. R. Kavitha	Mon-3, Fri-2	21	Registered
20CS203	Operating Systems	CSE	3	Dr. M. Suresh	Tue-2, Fri-1	1	Deny
20CS204	Computer Networks	CSE	3	Mr. K. Arun	Mon-1, Thu-3	30	Register
20CS205	Software Engineering	CSE	3	Ms. L. Divya	Wed-3, Fri-4	14	Registered
20CS206	Artificial Intelligence	AI&DS	4	Dr. V. Rajesh	Tue-4, Thu-2	0	Full
20CS207	Cyber Security	AI&DS	3	Mr. T. Harish	Wed-4, Fri-3	17	Registered
20CS208	Cloud Computing	IT	3	Ms. P. Anitha	Thu-4, Tue-3	15	Register
20MA201	Discrete Mathematics	MATHS	4	Dr. G. Latha	Mon-2, Wed-1	22	Register
20HS201	Professional Ethics	HSS	2	Mr. B. Karthik	Fri-5	30	Register
20CS279	Web Technologies Lab	CSE	2	Mr. N. Manoj	Thu-5	16	Register

Fig. 8 - Registering 20CS204 is rejected because it clashes with 20CS212 at Mon-1.

Course Registration Portal					
Welcome, Aishwarya U (71812401006)				Credits: 12 / 24	Logout
Available Courses My Registrations Timetable					
Credits registered: 12 / 24 Courses: 4					
Code	Course Title	Credits	Faculty	Slots	Action
20CS212	Web Technologies	3	Mr. N. Manoj	Mon-1, Wed-2	Drop
20CS202	Database Management	3	Ms. R. Kavitha	Mon-3, Fri-2	Drop
20CS205	Software Engineering	3	Ms. L. Divya	Wed-3, Fri-4	Drop
20CS207	Cyber Security	3	Mr. T. Harish	Wed-4, Fri-3	Drop

Fig. 9 - Four registered courses, 12 of 24 credits and a Drop button for each course.

Course Registration Portal					
Welcome, Aishwarya U (71812401006)				Credits: 12 / 24	Logout
Available Courses My Registrations Timetable					
Period	Mon	Tue	Wed	Thu	Fri
9:00 - 10:00	20CS212				
10:00 - 11:00			20CS212		20CS202
11:15 - 12:15	20CS202		20CS205		20CS207
1:15 - 2:15			20CS207		20CS205
2:15 - 3:15					

Fig. 10 - Weekly timetable with the slots of the registered courses highlighted.

Course Registration Portal

Welcome, Ajay Iyanraj (71812401007)

Credits: 0 / 24

Logout

Available Courses

My Registrations

Timetable

Search by code, title or faculty

All Departments

Code	Course Title	Dept	Credit
20CS212	Web Technologies	CSE	3
20CS201	Data Structures	CSE	4
20CS202	Database Management	CSE	3
20CS203	Operating Systems	CSE	3
20CS204	Computer Networks	CSE	3
20CS205	Software Engineering	CSE	3

Fig. 11 - The portal on a 390 px wide phone screen.

5. CONCLUSION

The Student Course Registration System was designed and implemented with HTML5, CSS3 and JavaScript. It allows a student to log in, search and filter the courses, register and drop courses and view the credits and the timetable. The rules for seat availability, credit limit and time clash are applied automatically, which removes the manual work and the mistakes of the paper based process. The test cases and the screenshots show that the application works correctly on desktop and mobile screens.

Challenges faced and solutions

Challenge	Solution
Keeping the seat count correct for all students	Seats left are calculated from all stored registrations every time the page is drawn
Finding a time clash between two courses	Each course stores its slots as Day-Period text and the slots are compared with find() and includes()
Data lost on page refresh	The registrations are saved in localStorage after every register or drop
Table too wide for a phone	A media query makes the table scroll horizontally on small screens

Learning outcomes

- Built a multi-view single page application using only HTML, CSS and JavaScript.
- Applied array methods (filter, find, reduce, map) to search and process the course data.
- Used localStorage to keep data in the browser and events to make the page interactive.

Future enhancements

- Connect the application to a server and a database (Node.js / MySQL) for real data storage.
- Add an administrator login to create courses, change seats and approve registrations.
- Send the registration confirmation by email and allow the timetable to be downloaded as a PDF.
- Add prerequisites, elective groups and a waiting list for full courses.

References

S. No.	Reference
1	MDN Web Docs - HTML, CSS and JavaScript reference (developer.mozilla.org)
2	MDN Web Docs - Window.localStorage and Web Storage API
3	W3Schools - JavaScript array methods and CSS grid tutorials